5. $\xi \sim R(\theta, 2\theta)$, $\theta > 0$.

а) $\vec{x}_n$.

Найти оценки $\theta$ (ОММ и ОМП)

• ОММ

$$\tilde{\alpha}_1 = \frac{1}{n}\sum x_i = \overline{x}$$

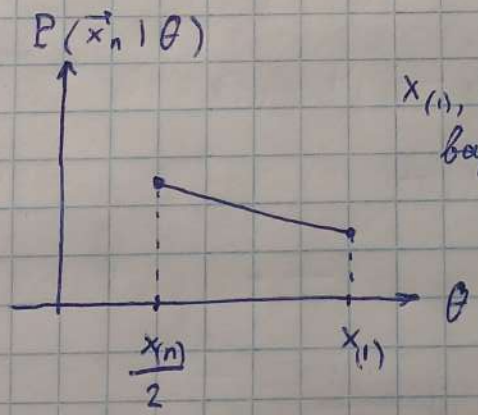
$$\alpha_1 = \int_{-\infty}^{\infty} x\, dF(x,\theta) = \int_{-\infty}^{\infty} x\, p(x,\theta)\, dx = \frac{1}{\theta}\int_{\theta}^{2\theta} x\, dx = \frac{3}{2}\theta.$$

$$p(x,\theta) = \frac{1}{2\theta-\theta}\{[\theta, 2\theta]\}$$

$\Rightarrow \tilde{\theta}_1 = \frac{2}{3}\overline{x}$ – ОММ.

• ОМП

$$P(\vec{x}_n|\theta) = \prod_{i=1}^{n} \frac{1}{\theta}\{[\theta, 2\theta]\}, \text{ т.е. } \begin{cases} \theta \le x_i & \forall i \\ 2\theta \ge x_i & \forall i, \end{cases} \text{ или же}$$




$x_{(1)}, x_{(n)}$ – эл. вар. ряда

$$\begin{cases} \theta \le x_{(1)} \\ \theta \ge \frac{x_{(n)}}{2} \end{cases} \Leftrightarrow \left[\frac{x_{(n)}}{2}, x_{(1)}\right] \quad B'$$

$$\underset{\theta \in B'}{\operatorname{argmax}} P(\vec{x}_n|\theta) = \frac{x_{(n)}}{2}$$

$\Rightarrow \tilde{\theta}_2 = \frac{x_{(n)}}{2}$ – ОМП.

б) Несмещенность и состоят. оценок.

$\cdot\, M\tilde{\theta}_1 = M\left(\frac{2}{3}\overline{X}\right) = \frac{2}{3\cancel{n}}\cancel{n}M\xi = \frac{2}{3}\cdot\frac{3}{2}\theta = \theta$ - несмещ.

$\cdot\, D\tilde{\theta}_1 = D\left(\frac{2}{3}\overline{X}\right) = \frac{4}{9n}D\xi = \frac{4}{9n}\left(M\xi^2 - (M\xi)^2\right) = \frac{4}{9n}\left(\int_\theta^{2\theta}\frac{x^2}{\theta}dx - \frac{9}{4}\theta^2\right) =$

$= \frac{4}{9n}\left(\frac{1}{3\cancel{\theta}}(8\theta^{\cancel{3}2} - \theta^{\cancel{3}2}) - \frac{9}{4}\theta^2\right) = \frac{4}{9n}\left(\frac{7}{3} - \frac{9}{4}\right)\theta^2 = \frac{4}{9n}\cdot\frac{1}{M_3}\theta^2 = \frac{\theta^2}{27n}$.

Т.о. для ОММ $\tilde{\theta}_1$ ~~все~~ $\forall\theta$ - несмещ. и $D\tilde{\theta}_1 \xrightarrow[n\to\infty]{} 0 \Rightarrow$

$\Rightarrow$ ОММ $\theta_1$ несмещ. и сост. ✓

$\cdot\, M\tilde{\theta}_2 = M\left(\frac{1}{2}X_{(n)}\right) = \frac{1}{2}M(X_{(n)}) = \frac{1}{2}\int_{-\infty}^{\infty} x\, dF_{(n)}(x,\theta) =$

$F_{(n)}(x) = F^n(x) \Rightarrow p(x,\theta) = nF^{n-1}(x) = \frac{n}{\theta}\left(\frac{x-\theta}{\theta}\right)^{n-1}\{[\theta, 2\theta]\}$

$= \frac{1}{2}\int_\theta^{2\theta} x\, d\left(\frac{x-\theta}{\theta}\right)^n = \frac{1}{2}\left[x\left(\frac{x-\theta}{\theta}\right)^n\Big|_\theta^{2\theta} - \frac{1}{\theta^n}\int_\theta^{2\theta}(x-\theta)^n d(x-\theta)\right] =$

$= \frac{1}{2}\left[2\theta - \frac{1}{(n+1)\theta^n}(x-\theta)^{n+1}\Big|_\theta^{2\theta}\right] \frac{1}{2}\left[2\theta - \frac{1}{n+1}\theta\right] =$

$= \frac{1}{2}\left[\frac{2n+\cancel{2}^{1}-\cancel{1}}{n+1}\right]\theta = \frac{2n+1}{2n+2}\theta$ - не явл. несмещ. ⊗

Исправл.: $\tilde{\theta}_2^* = \frac{\cancel{2}(n+1)}{2n+1}\cdot\frac{X_{(n)}}{\cancel{2}} = \frac{n+1}{2n+1}X_{(n)}$ - несмещ. ✓

$\cdot\, D\tilde{\theta}_2^* = \underbrace{\frac{(n+1)^2}{(2n+1)^2}}_{\gamma} D X_{(n)} = \gamma\left[M X_{(n)}^2 - M(X_n)^2\right] =$

$M X_{(n)}^2 = \int_\theta^{2\theta} x^2 d\left(\frac{x-\theta}{\theta}\right)^n = \frac{1}{\cancel{\theta^n}}\left[\cancel{\frac{x^2(x-\theta)^n}{\theta^n}}^{4\theta^2}\Big|_\theta^{2\theta} - \frac{2}{\cancel{\theta^n}}\int_\theta^{2\theta}\cancel{x}\left(\frac{x-\theta}{\theta}\right)^n dx\right] =$

$= \left[4\theta^2 - 2\int_\theta^{2\theta} x\left(\frac{x-\theta}{\theta}\right)^n d\frac{x-\theta}{\theta}\right] = \left[4\theta^2 - \frac{2\theta}{n+1}\int_\theta^{2\theta} x\, d\left(\frac{x-\theta}{\theta}\right)^{n+1}\right] =$

$= 4\theta^2 - \frac{2\theta}{n+1}\left[x\left(\frac{x-\theta}{\theta}\right)^{n+1}\Big|_\theta^{2\theta} - \frac{1}{(n+2)\theta^n}(x-\theta)^{n+2}\Big|_\theta^{2\theta}\right] \underset{n\to\infty}{\not\to} 0$.

~~Определение отсутств. состоят оценки~~

$\exists \theta > 0 \ \exists \varepsilon > 0 \quad P(|\tilde{\theta}_2^* - \theta| \geq \varepsilon) \underset{n\to\infty}{\not\to} 0$

$P(|\tilde{\theta}_2^* - \theta| \geq \varepsilon) = P\left(\frac{n+1}{2n+1} X_{(n)} \geq \varepsilon + \theta\right) + P\left(\frac{n+1}{2n+1} X_{(n)} \leq \theta - \varepsilon\right) =$

$= P\left(X_{(n)} \geq (\theta+\varepsilon)\frac{2n+1}{n+1}\right) + P\left(X_{(n)} \leq (\theta-\varepsilon)\frac{2n+1}{n+1}\right) \geq$

$\cancel{\Rightarrow P\left(X_{(n)} \leq (\theta-\varepsilon)\frac{2n+1}{n+1}\right) \ \exists \varepsilon \geq \theta} \geq P\left(X_{(n)} < (\theta-\varepsilon)\frac{2n+1}{n+1}\right) = \left\{\varepsilon < \frac{\theta}{2}\right\}$

$= F_{(n)}\left((\theta-\varepsilon)\frac{2n+1}{n+1}\right) = \left(\frac{\theta\frac{2n+1}{n+1} - \theta\frac{n+1}{n+1} - \varepsilon\frac{2n+1}{n+1}}{\theta}\right)^n =$

$= \left(\frac{n}{n+1} - \frac{\varepsilon}{\theta}\frac{2n+1}{n+1}\right)^n$

Определение состоят. оценки:

$\forall \theta > 0 \ \forall \varepsilon > 0 \quad P(|\tilde{\theta}_2^* - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$

$P(|\tilde{\theta}_2^* - \theta| \geq \varepsilon) = P\left(\frac{n+1}{2n+1} X_{(n)} \geq \varepsilon + \theta\right) + P\left(\frac{n+1}{2n+1} X_{(n)} \leq \theta - \varepsilon\right) =$

$= P\left(X_{max} \geq (\theta+\varepsilon)\left(2 - \frac{1}{n+1}\right)\right) + P\left(X_{max} \leq (\theta-\varepsilon)\left(2 - \frac{1}{n+1}\right)\right) = 0$

При: $2\theta < (\theta+\varepsilon)\left(2 - \frac{1}{n+1}\right)$

$0 < 2\varepsilon - \frac{\theta\cdot\varepsilon}{n+1}$

$n_1 > \frac{\theta+\varepsilon}{2\varepsilon} - 1 = \frac{\theta-\varepsilon}{2\varepsilon}$

$\theta > (\theta-\varepsilon)\left(2 - \frac{1}{n+1}\right)$

$0 > \theta - 2\varepsilon - \frac{\theta-\varepsilon}{n+1}$

$2\varepsilon - \theta > -\frac{\theta-\varepsilon}{n+1}$

$n+1 > \frac{\theta-\varepsilon}{\theta-2\varepsilon}$

$n_2 > \frac{\varepsilon}{\theta-2\varepsilon}$

$n > \max(n_1, n_2) \Rightarrow$

$\Rightarrow$ оценка состоятельна!

Таким образом, ОМП $\tilde{\theta}_2^*$ несмещ. и состоят. ✓

c) Исследование эффективн. оценок

. Регулярн. модели:

1) $F(x,\theta) = \frac{x-\theta}{\theta}$ – непрер. диф. по $\theta$, при $\theta > 0$.

2) $\int_{\theta}^{2\theta} d\left(\frac{-\theta-(x-\theta)}{\theta^2}\right) = \int_{\theta}^{2\theta} d\left(-\frac{x}{\theta^2}\right) = \int_{\theta}^{2\theta} -\frac{1}{\theta^2}\,dx = -\frac{1}{\theta} \neq 0$

$\Rightarrow$ модель не явл. регулярной.

$\Rightarrow$ нерав-во Крамера-Рао неприменимо.

$D\tilde{\theta}_1 = \frac{\theta^2}{27n} \xrightarrow[n\to\infty]{} 0$

$D\tilde{\theta}_2^* = \left(\frac{n+1}{2n+1}\right)^2 \left[4\theta^2 - \frac{4\theta^2}{n+1} + \frac{2\theta^2}{(n+2)(n+1)}\right] \underset{n\to\infty}{\not\to} 0$

$\Rightarrow$ Начиная с какого-то $n$ $\tilde{\theta}_1$ ближе к эфф. оценке в смысле дисперсии, чем $\tilde{\theta}_2^*$.

d) точный доверительный интервал для $\theta$.

$\frac{X_i}{\theta} \sim R(1,2) \to \underbrace{\frac{X_{(n)}}{\theta}}_{f(\vec{x}_n,\theta)} \sim \left(\frac{t-1}{2-1}\right)^n$

$\left(\frac{t-1}{1}\right)^n = \frac{1-\beta}{2} \Rightarrow t_{\frac{1-\beta}{2}} = 1+\sqrt[n]{\frac{1-\beta}{2}}$ $\qquad t_{\frac{1+\beta}{2}} = 1+\sqrt[n]{\frac{1+\beta}{2}}$

$1-\sqrt[n]{\frac{1-\beta}{2}} < \frac{X_{(n)}}{\theta} < 1+\sqrt[n]{\frac{1+\beta}{2}} \Rightarrow \frac{X_{(n)}}{1-\sqrt[n]{\frac{1-\beta}{2}}} < \theta < \frac{X_{(n)}}{1+\sqrt[n]{\frac{1-\beta}{2}}}$ – довер. инт. для $\theta$.

е) ЦПТ: $\dfrac{\overline{X} - \frac{3}{2}\theta}{\frac{1}{12}\theta^2}\sqrt{n} \sim N(0,1)$

$$t_{\frac{1-\beta}{2}} < \frac{\overline{X} - \frac{3}{2}\theta}{\frac{1}{12}\theta^2}\sqrt{n} < t_{\frac{1-\beta}{2}}$$

$$t_{\frac{1-\beta}{2}} < \frac{\overline{X} - \frac{3}{2}\theta}{S_2}\sqrt{n} < t_{\frac{1-\beta}{2}}$$

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$$\frac{2}{3}\left(\overline{X} - t_{\frac{1+\beta}{2}}\frac{S_2}{\sqrt{n}}\right) < \theta < \frac{2}{3}\left(\overline{X} - t_{\frac{1-\beta}{2}}\frac{S_2}{\sqrt{n}}\right)$$

– сост. оценка дисперсии

асимпт. довер. интерв